Measure Theoretic Probability
Lesson 18
$\sigma(X)$ for Random Vectors and Almost Sure Convergence

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Outline

1. Recap: Random Variables and Random Vectors
2. Structure of $\sigma(X)$
3. Relating $\sigma(X_1, \ldots, X_k)$ to the $\sigma(X_i)$
4. Limits of Random Variables
5. Summary and Exercises

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Random Variables and Random Vectors

Random variable

$(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space and $\mathbb{R}$ carries the Borel σ-field $\mathcal{B}(\mathbb{R})$. A function $X : \Omega \to \mathbb{R}$ is a random variable if it is measurable: $X^{-1}(B) \in \mathcal{F}$ for every $B \in \mathcal{B}(\mathbb{R})$. (No measure is needed for this condition.)

Random vector

$\mathsf{X} = (X_1, \ldots, X_k) : \Omega \to \mathbb{R}^k$, with $\mathsf{X}(\omega) = \big(X_1(\omega), \ldots, X_k(\omega)\big)$, is a random vector if it is measurable $\mathcal{F}/\mathcal{B}(\mathbb{R}^k)$.

Result from Lesson 17

A vector-valued function is measurable if and only if each of its components is measurable. So X is a random vector $\iff$ each X_i is a random variable.

The σ-Field Generated by a Random Vector

Definition (vector form)

$\sigma(\mathbf{X}) = \sigma\big((X_1, \ldots, X_k)\big)$ is the **smallest** σ-field on Ω with respect to which the vector $\mathbf{X}$ is measurable: all sets $\mathbf{X}^{-1}(B)$, $B \in \mathcal{B}(\mathbb{R}^k)$, lie in it.

Definition (component form)

$\sigma(X_1, \ldots, X_k)$ is the **smallest** σ-field with respect to which each component X_i is measurable: all sets ${X_i}^{-1}(B)$, $B \in \mathcal{B}(\mathbb{R})$, lie in it.

Notation

Double parentheses are the vector form; single parentheses are the component form. They are the **same** σ-field (shown for simple variables earlier; the general proof is similar and is omitted). From now on we do not distinguish the two.

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Theorem (Part I and Part II)

Theorem

Let $\mathrm{X} = (X_1, \ldots, X_k)$ be a k-dimensional random vector on $(\Omega, \mathcal{F}, \mathbb{P})$.

(I) $\sigma(\mathrm{X})$ consists of all inverse images of Borel sets of $\mathbb{R}^k$:

$$\sigma(\mathrm{X}) = \left\{\mathrm{X}^{-1}(B) : \ B \in \mathcal{B}(\mathbb{R}^k)\right\}.$$

(II) A random variable Y is $\sigma(\mathrm{X})$-measurable if and only if $Y = f(\mathrm{X})$ for some **measurable** function $f : \mathbb{R}^k \to \mathbb{R}$.

Remarks

- Taking $k = 1$ gives the one-dimensional theorem stated in Lesson 17.
- Today we prove (I). Part (II) is postponed for another lesson. Its proof uses limits of sequences of random variables.
- Equivalent notations: $\mathrm{X}^{-1}(B) = (X_1, \ldots, X_k)^{-1}(B) = \{\mathrm{X} \in B\}$. We will move freely between them.

Proof of (I): Strategy

Define the candidate class

$$\mathcal{M} = \{\mathsf{X}^{-1}(B) : \ B \in \mathcal{B}(\mathbb{R}^k)\}.$$

We must show $\mathcal{M} = \sigma(\mathsf{X})$.

Plan

1. $\mathcal{M} \subseteq \sigma(\mathsf{X})$: every pullback of a Borel set is in $\sigma(\mathsf{X})$ by definition.
2. $\mathcal{M}$ is a σ-field.
3. Each X_i is $\mathcal{M}$-measurable.
4. Conclude by minimality.

Why no "filling in" is needed

Complements and unions of Borel sets are again Borel sets, and pulling back commutes with complements and unions. So pullbacks of Borel sets are already closed under these operations.

Step 2: $\mathcal{M}$ Is Closed Under Complements

Take $A \in \mathcal{M}$

By definition of $\mathcal{M}$, $A = \mathrm{X}^{-1}(B) = \{\omega : \mathrm{X}(\omega) \in B\}$ for some $B \in \mathcal{B}(\mathbb{R}^k)$.

Complement

Every point of A maps into B. Every point outside A maps outside B: if it mapped into B, it would belong to $A = \mathrm{X}^{-1}(B)$. Note that X need not be onto $\mathbb{R}^k$, so not every point of B is hit. Hence

$$A^c = \{\omega : \mathrm{X}(\omega) \in B^c\} = \mathrm{X}^{-1}(B^c).$$

$B^c \in \mathcal{B}(\mathbb{R}^k)$ since $\mathcal{B}(\mathbb{R}^k)$ is a σ-field, so $A^c \in \mathcal{M}$.

Remaining properties

$\Omega = \mathrm{X}^{-1}(\mathbb{R}^k) \in \mathcal{M}$, and closure under countable unions follows from $\bigcup_j \mathrm{X}^{-1}(B_j) = \mathrm{X}^{-1}\left(\bigcup_j B_j\right)$. These are left as routine exercises.

Step 3: Each X_i Is $\mathcal{M}$-Measurable

Pull back a one-dimensional Borel set

Let $B \in \mathcal{B}(\mathbb{R})$. The points of Ω that X_i sends into B are exactly the points that X sends into the set with B in the i-th coordinate and no restriction elsewhere:

$$X_i^{-1}(B) = \mathrm{X}^{-1}\big(\mathbb{R} \times \cdots \times \mathbb{R} \times B \times \mathbb{R} \times \cdots \times \mathbb{R}\big),$$

with k factors and B in position i.

Is that set Borel in $\mathbb{R}^k$?

Yes. $\mathbb{R} \in \mathcal{B}(\mathbb{R})$ and $B \in \mathcal{B}(\mathbb{R})$, so the product set lies in

$$\mathcal{B}(\mathbb{R}) \times \cdots \times \mathcal{B}(\mathbb{R}) \;\subseteq\; \sigma\big(\mathcal{B}(\mathbb{R}) \times \cdots \times \mathcal{B}(\mathbb{R})\big) = \mathcal{B}(\mathbb{R}^k),$$

using the result of Lesson 17. Hence $X_i^{-1}(B) \in \mathcal{M}$.

Step 4: Conclusion of the Proof of (I)

Recap

- $\mathcal{M} \subseteq \sigma(\mathrm{X})$ (by definition of $\sigma(\mathrm{X})$);
- $\mathcal{M}$ is a σ-field;
- each X_i is $\mathcal{M}$-measurable.

Minimality

$\sigma(\mathrm{X})$ is the **smallest** σ-field for which each X_i is measurable. Since $\mathcal{M}$ is such a σ-field, $\sigma(\mathrm{X}) \subseteq \mathcal{M}$. Together with $\mathcal{M} \subseteq \sigma(\mathrm{X})$,

$$\mathcal{M} = \sigma(\mathrm{X}). \qquad \square$$

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Theorem: Joint σ-Field from the Marginals

Theorem

For random variables $X_1, \ldots, X_k$ on $(\Omega, \mathcal{F}, \mathbb{P})$,

$$\sigma(X_1, \ldots, X_k) = \sigma\big(\sigma(X_1) \cup \cdots \cup \sigma(X_k)\big).$$

Scope of the proof

We prove $k = 2$. The general case follows by induction: the union of the first two fields is handled as above, then adjoin the third, and so on.

Two inclusions

($\subseteq$) $\sigma(X_1, X_2) \subseteq \sigma\big(\sigma(X_1) \cup \sigma(X_2)\big)$.

($\supseteq$) $\sigma\big(\sigma(X_1) \cup \sigma(X_2)\big) \subseteq \sigma(X_1, X_2)$.

Proof of ($\subseteq$): Reduce to Generating Sets

Why not start from a set in $\sigma(X_1, X_2)$?

Taking an arbitrary set of the left side and showing it lies on the right is not fruitful. Instead we use part (I): a set in $\sigma(X_1, X_2)$ is $(X_1, X_2)^{-1}(B)$ with $B \in \mathcal{B}(\mathbb{R}^2)$.

Use the Lemma (Lesson 17)

It is enough to pull back a **generating class** of $\mathcal{B}(\mathbb{R}^2)$. Two natural choices are rectangles $(a_1, a_2) \times (b_1, b_2)$, or products $A \times B$ with $A, B \in \mathcal{B}(\mathbb{R})$. We use the second, since $\mathcal{B}(\mathbb{R}^2) = \sigma\big(\mathcal{B}(\mathbb{R}) \times \mathcal{B}(\mathbb{R})\big)$.

Goal

Show $(X_1, X_2)^{-1}(A \times B) \in \sigma\big(\sigma(X_1) \cup \sigma(X_2)\big)$ for all Borel $A, B \subseteq \mathbb{R}$. By the Lemma, every set of $\sigma(X_1, X_2)$ will then lie there.

Proof of ($\subseteq$): The Key Computation

Pullback of a product set

Let $A, B \in \mathcal{B}(\mathbb{R})$. A point ω is sent into $A \times B$ exactly when $X_1(\omega) \in A$ **and** $X_2(\omega) \in B$:

$$(X_1, X_2)^{-1}(A \times B) = \{X_1 \in A\} \cap \{X_2 \in B\} = {X_1}^{-1}(A) \cap {X_2}^{-1}(B).$$

Locate the two sets

${X_1}^{-1}(A) \in \sigma(X_1)$ and ${X_2}^{-1}(B) \in \sigma(X_2)$, by part (I) in one dimension. So both lie in the "bucket" $\sigma(X_1) \cup \sigma(X_2)$.

Intersections are in the generated σ-field

The bucket need not be closed under intersections, but $\sigma\big(\sigma(X_1) \cup \sigma(X_2)\big)$ is a σ-field containing it. So the intersection lies in $\sigma\big(\sigma(X_1) \cup \sigma(X_2)\big)$. By the Lemma, $\sigma(X_1, X_2) \subseteq \sigma\big(\sigma(X_1) \cup \sigma(X_2)\big)$. □

Proof of $(\supseteq)$

Step 1: $\sigma(X_1) \subseteq \sigma(X_1, X_2)$

Let $A \in \sigma(X_1)$. By part (I) in one dimension, $A = {X_1}^{-1}(B)$ for some $B \in \mathcal{B}(\mathbb{R})$. Then

$$A = (X_1, X_2)^{-1}(B \times \mathbb{R}),$$

since X_2 is unrestricted. $B \times \mathbb{R} \in \mathcal{B}(\mathbb{R}) \times \mathcal{B}(\mathbb{R}) \subseteq \mathcal{B}(\mathbb{R}^2)$, because $\mathbb{R} \in \mathcal{B}(\mathbb{R})$. So $A \in \sigma(X_1, X_2)$. The argument for $\sigma(X_2)$ is the same.

Step 2: the union and its generated σ-field

Step 1 gives $\sigma(X_1) \cup \sigma(X_2) \subseteq \sigma(X_1, X_2)$. Now $\sigma(X_1, X_2)$ is a σ-field containing this union, and $\sigma\big(\sigma(X_1) \cup \sigma(X_2)\big)$ is the smallest such σ-field. Hence

$$\sigma\big(\sigma(X_1) \cup \sigma(X_2)\big) \subseteq \sigma(X_1, X_2). \qquad \square$$

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What Does $X_n \to X$ Mean?

Setting

Let $X_1, X_2, X_3, \ldots$ be a sequence of random variables on a **common** probability space $(\Omega, \mathcal{F}, \mathbb{P})$. A common space is needed so that all pullbacks lie in the same σ-field $\mathcal{F}$.

The difficulty

Random variables are functions, not numbers, so the statement "$\lim_{n\to\infty} X_n = X$" needs a definition. We must bring in probability to produce numbers whose limits we know how to take.

Two ways to produce numbers

- **Expected values:** $\mathbb{E}[X_n] \to \mathbb{E}[X]$ (a limit of real numbers). Expectation has not been covered yet.
- **Probabilities:** for every $\varepsilon > 0$, $\mathbb{P}(|X_n - X| > \varepsilon) \to 0$ as $n \to \infty$. This is **convergence in probability**.

Convergence in Probability, as Sets

Measure-theoretic form

The probability $\mathbb{P}(|X_n - X| > \varepsilon)$ is the measure of a set of outcomes:

$$\mathbb{P}\big(\{\omega \in \Omega : \ |X_n(\omega) - X(\omega)| > \varepsilon\}\big) \longrightarrow 0 \qquad (n \to \infty)$$

for every $\varepsilon > 0$.

Remark

In mathematical statistics this is usually written with the abbreviated notation. In measure theory everything is a statement about sets of ω's.

Almost Sure Convergence

Definition

X_n **converges to** X **almost surely** if

$$\mathbb{P}\Big(\big\{\omega \in \Omega : \lim_{n\to\infty} X_n(\omega) = X(\omega)\big\}\Big) = 1.$$

We write $X_n \xrightarrow{\text{a.s.}} X$.

What is happening

Fix ω. The values $X_1(\omega), X_2(\omega), \ldots$ form a sequence of **real numbers**, so the limit makes sense. Almost sure convergence says that this sequence converges to $X(\omega)$ for all ω, or at least for all ω outside a set of probability zero.

Other names

This is the same as **almost everywhere** convergence in general measure theory and analysis, and it is also called **convergence with probability one**.

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Summary

Key results

Concept / Result	Content
Random vector	$\mathrm{X} : \Omega \to \mathbb{R}^k$ measurable $\iff$ each X_i measurable
$\sigma(\mathrm{X})$	smallest σ-field making X (equivalently each X_i) measurable
Theorem (I)	$\sigma(\mathrm{X}) = \{\mathrm{X}^{-1}(B) : \ B \in \mathcal{B}(\mathbb{R}^k)\}$
Theorem (II)	Y is $\sigma(\mathrm{X})$-measurable $\iff$ $Y = f(\mathrm{X})$, f measurable
Joint vs marginals	$\sigma(X_1, \ldots, X_k) = \sigma\big(\sigma(X_1) \cup \cdots \cup \sigma(X_k)\big)$
Almost sure convergence	$\mathbb{P}\big(\lim_n X_n = X\big) = 1$

Exercises

1. Show that $\mathcal{M} = \{X^{-1}(B) : \ B \in \mathcal{B}(\mathbb{R}^k)\}$ contains Ω and is closed under countable unions.
2. Show that $(X_1, X_2)^{-1}(A \times B) = {X_1}^{-1}(A) \cap {X_2}^{-1}(B)$.
3. Show that $\sigma(X_2) \subseteq \sigma(X_1, X_2)$ by writing $A \in \sigma(X_2)$ as a pullback under (X_1, X_2) of $\mathbb{R} \times B$.
4. Explain why $\sigma(X_1) \cup \sigma(X_2)$ need not be closed under intersections, and why $\sigma\big(\sigma(X_1) \cup \sigma(X_2)\big)$ fixes this.
5. Prove the joint-versus-marginals theorem for $k = 3$ by induction from the case $k = 2$.
6. Let $X_n(\omega) = \omega^n$ on $\Omega = [0, 1]$ with Lebesgue measure. Find $\{\omega : \ \lim_n X_n(\omega) \text{ exists}\}$.

Outlook

Next lesson

An equivalent characterization of almost sure convergence that is easier to use, followed by examples. Then convergence in probability, and the fact that almost sure convergence implies convergence in probability, but not conversely.

Looking ahead

Part (II) of the $\sigma(X)$ theorem, that measurability with respect to $\sigma(X)$ means being a measurable function of X, will be proved once limits of random variables are available.